Automated Gastric Cancer Diagnosis from Endoscopic Images

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***Abstract*—Detection of gastrointestinal (GI) abnormalities in endoscopic images is a necessary step before targeted biopsy of early-stage gastric ulcers and carcinomas. Although deep networks perform well on public benchmarks such as Kvasir, real-world deployment is hindered by real-time constraints, strong class imbalance and transient camera artifacts that cause false positives. This paper introduces a cascaded spatio-temporal framework: a lightweight MobileNetV2 gate filters anatomically normal frames, so that a ResNet-50 feature extractor and a Bi-directional Long Short-Term Memory (Bi-LSTM) network are run only on suspicious frames, where the Bi-LSTM verifies lesion stability across a sliding window of $k = 5$ frames. Focal loss handles class imbalance and Grad-CAM supplies visual explanations. The system is benchmarked against a classical SIFT bag-of-visual-words + SVM pipeline and eight ImageNet-pretrained CNN/Transformer baselines (AlexNet, VGG-16, GoogLeNet, ResNet-50, DenseNet-121, MobileNetV2, EfficientNet-B0, ViT-B/16) on *four* public databases (Kvasir v2, GastroVision, Kvasir-SEG, CVC-ClinicDB) using ten performance evaluators. On Kvasir v2 the proposed cascade reaches 91.92% accuracy and an MCC of 90.79%, 1.7 points below the best single-frame baseline (ViT-B/16, 93.58%). Its advantage appears on degraded frames: when the current frame is hit by a transient artifact, it keeps 91.8% accuracy versus 74.1% for the most robust single-frame baseline. The temporally smoothed gate passes 99.3% of abnormal frames while normal frames skip the expensive stages, lowering the average cost to 3.35 GMACs per frame versus 4.09 for ResNet-50.**

***Index Terms*—Gastrointestinal cancer, endoscopy, computer-aided detection, deep learning, Bi-LSTM, focal loss, Grad-CAM, Kvasir.**

I. Introduction & Relevance

Malignancies of the gastrointestinal tract represent a considerable burden to world health. Their clinical outcome depends heavily on the stage at which diagnosis occurs; screening for early-stage gastric ulcers and carcinomas is therefore of particular importance to reduce mortality and avoid radical treatment. The clinical standard for GI examination is endoscopy, which offers real-time, high-resolution visualisation of the mucosa of the oesophagus, stomach and colon. However, detecting subtle, flat pre-cancerous lesions by eye limits the endoscopist's ability to perform targeted biopsies in real time, as diagnostic performance depends heavily on skill and fatigue.

Traditional computer vision approaches based on hand-crafted descriptors such as SIFT [8], corner and edge detectors often fail to robustly describe the homogeneous, highly specular textures of the GI tract. Convolutional neural networks that learn hierarchical representations directly from pixels bypass the geometric and photometric constraints of traditional image processing, and have become the dominant approach.

Automation should reduce the workload and fatigue of clinicians, but it must be designed around practical concerns: false negatives cost lives, while false positives lead to unnecessary invasive biopsies and cost. The contributions of this work are:

1) A three-stage *cascaded spatio-temporal* pipeline (MobileNetV2 gate $\rightarrow$ ResNet-50 features $\rightarrow$ Bi-LSTM) whose gate threshold is chosen to retain $\geq$99.5% of abnormal frames while letting most normal frames skip the expensive stages.
2) A synthetic video protocol that converts still-image databases into 5-frame clips with simulated camera motion and four families of endoscopic artifacts, enabling a controlled study of temporal robustness.
3) A broad baseline comparison of nine methods on four public databases with ten performance evaluators, including efficiency (parameters, GMACs, FPS), cross-database generalisation and quantitative Grad-CAM localisation.

II. Literature Review & Core Research Gaps

AI-based technologies in gastroenterology have been greatly facilitated by growth in computation, deep learning methods and publicly available datasets. The Kvasir database [1] was a milestone: a unified multi-class benchmark spanning anatomical landmarks, pathological findings and endoscopic procedures, it accelerated the design of computer-aided detection (CAD) systems. It was followed by larger collections such as HyperKvasir [2] and GastroVision [3], and by pixel-annotated polyp sets such as Kvasir-SEG [4] and CVC-ClinicDB [5]. Systematic reviews and meta-analyses have provided convincing evidence for the superiority of CNNs over conventional machine learning for detecting gastric cancer and colorectal lesions [6], [7].

A. Architectural Evolution in Endoscopic Vision

Early GI-lesion classifiers reused frameworks developed for natural images [12]. VGG [13] and Inception/GoogLeNet

[14], [15] improved multi-scale feature extraction, but their parameter counts, memory requirements and slow inference constrain clinical use. Residual connections in ResNet [16] and dense feature concatenation in DenseNet [17] enabled much deeper networks with better robustness on Kvasir [18]; for example, ResNet-152 with class activation mapping exceeded 93% accuracy while localising discriminative regions [18], [19].

Beyond whole-image classification, U-Net [20] dominates lesion segmentation, while two-stage detectors such as Faster R-CNN [21] and Mask R-CNN [22] provide accurate box and instance detection of polyps and ulcers, though often at the cost of processing speed.

B. Real-Time Processing and Ensemble Techniques

Because real-time feedback guides the endoscopist's instruments, efficient architectures such as MobileNetV2 [23] and EfficientNet [24] use depthwise-separable convolutions and compound scaling to deliver rich features at low cost. Hybrid ensembles inject global context from heavier networks [25], and single-stage detectors such as YOLOv4 [26] remove the region-proposal bottleneck. Optimising such models on skewed medical distributions requires the Adam family of optimisers [27], [28] and re-weighted losses such as Focal Loss [30]. Vision Transformers [31] promise better global context but their cost and latency are a concern for real-time endoscopy.

C. Core Research Gaps

Despite these advances, three gaps hinder the translation of static benchmark accuracy into dynamic clinical use:

1) **Throughput bottlenecks during live inference:** ensembles and Transformers achieve high accuracy on static images, but their cost makes it hard to meet strict latency/FPS requirements.
2) **Class imbalance and under-represented rare pathologies:** cross-entropy is biased towards majority classes (e.g. normal tissue); early flat pre-cancerous lesions remain under-represented in datasets such as Kvasir.
3) **Lack of clinical explainability:** high-accuracy classifiers seldom give fine-grained spatial justification, and Grad-CAM heat-maps are rarely *validated* against expert lesion annotations.

In addition, almost all published systems classify frames independently and are thus vulnerable to transient artifacts (motion blur, specular flashes, bubbles) that are ubiquitous in live video.

D. Comparison with Baseline Architectures & State-of-the-Art

Table I summarises the trade-offs reported in the literature, together with the results of the proposed system measured in this work (Section VII).

III. Databases and Materials

Four public databases are used (Table II, Fig. 1). **Kvasir v2** [1] is the primary benchmark: 8,000 expert-annotated images, balanced across eight classes of anatomical landmarks (Z-line, pylorus, cecum), pathological findings (oesophagitis, polyps, ulcerative colitis) and procedures (dyed-lifted polyps, dyed resection margins). **GastroVision** [3] is a strongly imbalanced multi-class database covering the upper and lower GI tract, including gastric findings such as gastric polyps, ulcers, erythema and colorectal cancer. Classes with fewer than 30 images (7 in GastroVision) are removed, because they cannot be split into train/validation/test. **Kvasir-SEG** [4] and **CVC-ClinicDB** [5] contain polyp frames with pixel-level expert masks; they are never used for training and serve as external test sets for cross-database generalisation and quantitative Grad-CAM localisation. CVC-ClinicDB was acquired with a different endoscope at a different hospital, making it a genuine domain-shift test.

Each classification database is split once, stratified by class, into 70/15/15% train/validation/test sets (seed 42). Raw images (720×576 to 1920×1072 pixels) are resized with bicubic interpolation to $224\times224\times3$ and cached as 8-bit tensors, so high-resolution JPEGs are decoded only once.

The pipeline is implemented in PyTorch 2 with torchvision ImageNet weights and runs on an Apple M5 GPU (Metal/MPS backend) with mixed-precision (FP16) arithmetic; augmentation, artifact simulation and clip synthesis are executed on the GPU to avoid data-loading bottlenecks. All code, configuration and logs are released with the paper.

IV. Proposed System Design

Considering the clinical constraints of high-throughput endoscopy and the imbalanced nature of GI disease, the system is designed as a cascaded, spatio-temporal diagnosis pipeline (Fig. 3).

A. Synthetic Clips from Still Images

To emulate the time-series nature of clinical video from still-image databases, each image I is turned into a $k = 5$-frame clip $\{I_{t-4}, \ldots, I_t\}$ by a smooth simulated camera trajectory: a per-clip random velocity in rotation ($\leq 4°$/frame), zoom ($\leq 3\%$/frame) and translation ($\leq 3\%$/frame) is integrated backwards from the current frame I_t. Any frame may then be replaced by a corrupted version produced by one of four artifact models (Fig. 2): *motion blur* (random line kernel, 11–23 px), *defocus* (Gaussian, $\sigma \in [2.5, 6]$), *specular flash* (global overexposure plus saturated specular blobs) and *noise/occlusion* (Gaussian sensor noise and a dark occluding region, e.g. a bubble or debris).

B. Spatio-Temporal Architectural Flow

Stage 1 – gate. A MobileNetV2 classifier is trained over all classes; its binary gate score is the total probability of the abnormal classes $\mathcal{A}$ (pathology and interventions), $s_t = \sum_{c\in\mathcal{A}} P(c \mid I_t)$. Its efficiency comes from depthwise-separable convolution, which factorises a standard convolution

TABLE I

PERFORMANCE TRADE-OFFS OF ENDOSCOPIC IMAGE ANALYSIS APPROACHES (LITERATURE VALUES ARE TYPICAL FIGURES REPORTED ON KVASIR-TYPE DATA; THE LAST ROW IS MEASURED IN THIS WORK).

Reference Framework	**Analysis Modality**	**Primary Optimization Focus**	**Typical Accuracy**	**Real-Time Efficiency**
VGG-16 / Inception [13], [15]	Static image	Multi-scale feature extraction	∼85–88%	Low FPS
ResNet-152 + CAM [18]	Static image	Discriminative lesion localisation	>93%	Medium FPS
MobileNetV2 / EfficientNet [23], [24]	Static image	High-throughput processing	∼90%	High FPS
YOLOv4 [26]	Static detection	Rapid bounding-box generation	∼91%	Very high FPS
Proposed system	**Spatio-temporal**	**Gated sequential lesion tracking**	**91.92%**	**3.35 GMACs/frame**

TABLE II

DATABASES USED IN THIS STUDY (ALL PUBLIC). IMBALANCE = LARGEST/SMALLEST CLASS SIZE.

Database	Images	Classes	Imbalance	Train / Val / Test	Role in this study
Kvasir v2	8,000	8	1.0	5600 / 1200 / 1200	8-class train/test (primary)
GastroVision	7,872	20	22.6	5510 / 1181 / 1181	multi-class train/test (upper+lower GI)
Kvasir-SEG	1,000	1 (+mask)	–	test only	external polyp test + localisation
CVC-ClinicDB	612	1 (+mask)	–	test only	external polyp test + localisation

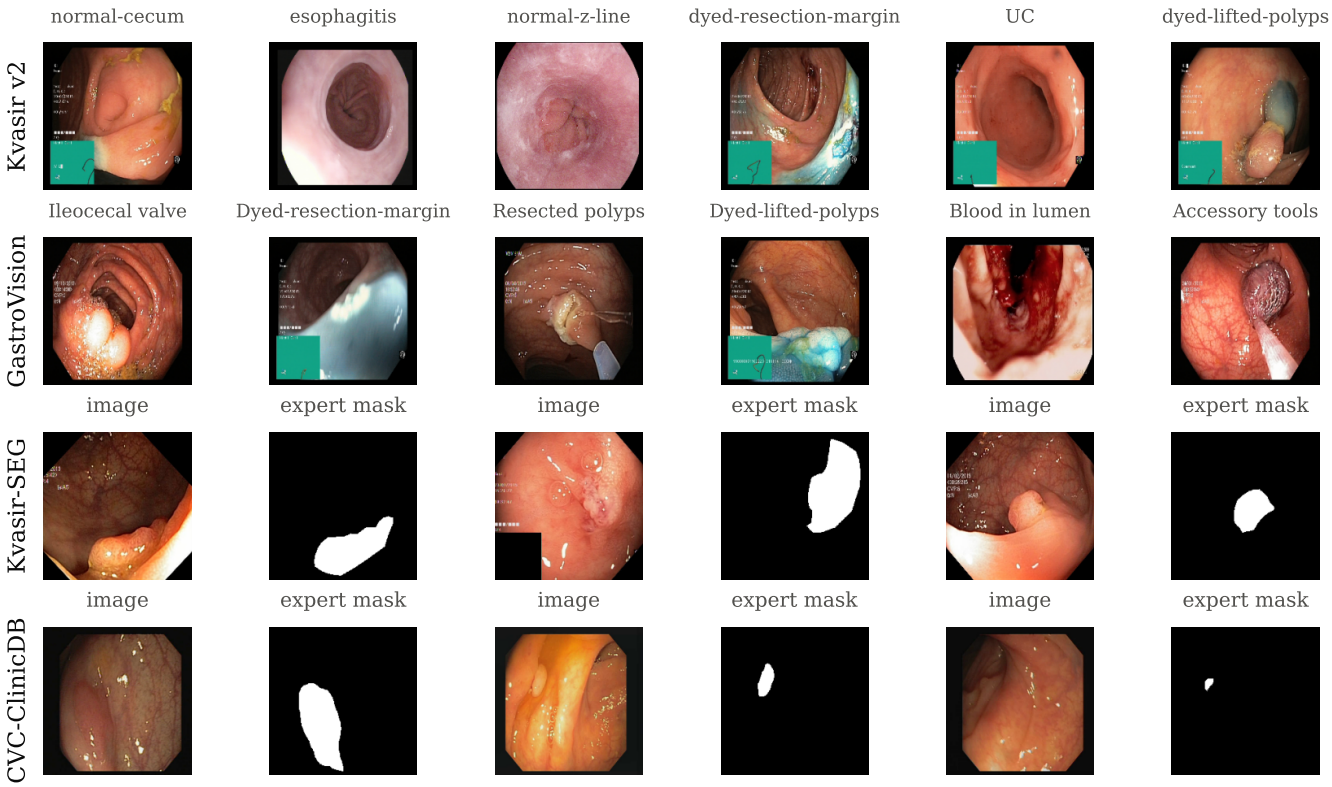


Fig. 1. Samples from the four databases: random classes from Kvasir v2 and GastroVision, and image/mask pairs from the two external polyp databases.

into a depthwise spatial convolution and a 1×1 pointwise convolution. For an input feature map F and depthwise kernel $\hat{K}$, the depthwise output is

$$\hat{G}_{k,l,m} = \sum_{i,j} \hat{K}_{i,j,m} \cdot F_{k+i-1,l+j-1,m}, \tag{1}$$

followed by a pointwise linear combination with a 1×1 kernel P:

$$G_{k,l,n} = \sum_{m} P_{m,n} \cdot \hat{G}_{k,l,m}. \tag{2}$$

Frames with $s_t < \tau$ are labelled with the gate's most probable *normal* class and skip the deeper stages, which reduces the average latency.

Stage 2 – spatial features. Frames with $s_t \geq \tau$ are passed

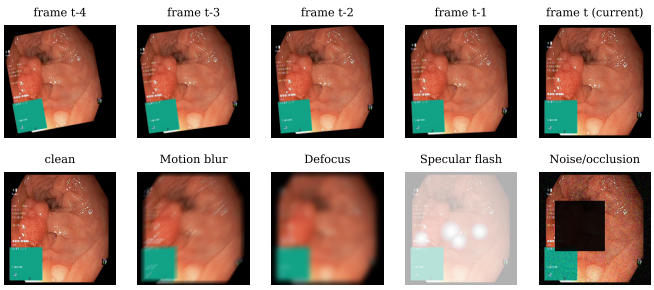


Fig. 2. Top: a synthetic 5-frame clip generated from one Kvasir polyp image. Bottom: the four simulated artifact families applied to the current frame.

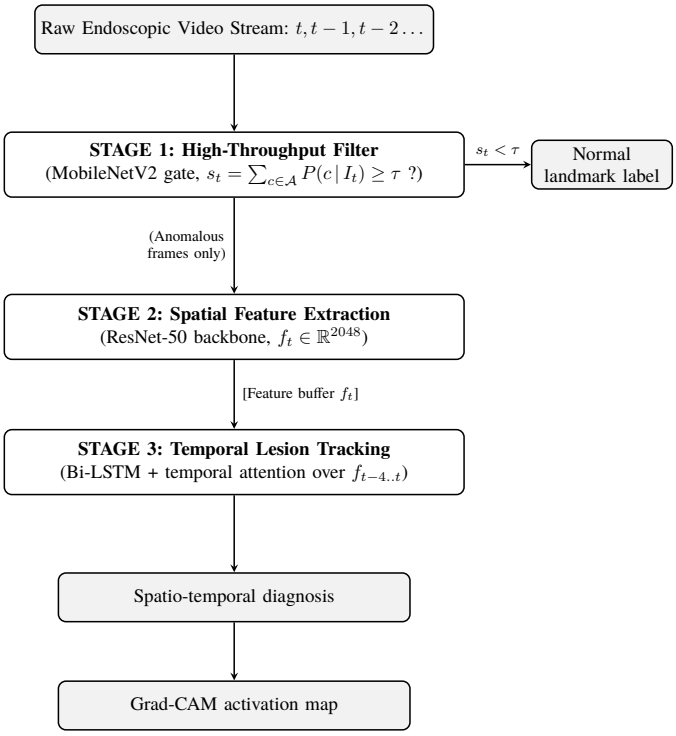


Fig. 3. Proposed spatio-temporal cascaded architecture. Frames whose abnormality score s_t is below τ exit after Stage 1 with a normal-landmark label.

to a ResNet-50 whose global-pooled output is a 2048-d spatial feature vector f_t, pushed into a first-in-first-out buffer of the last k features.

Stage 3 – temporal verification. To suppress false positives caused by transient blur or light flashes, a Bi-LSTM [32], [33] processes the buffer. After a LayerNorm-and-linear projection to 512-d, it computes forward and backward hidden states (256 units each) at every time step:

$$\begin{aligned} \overrightarrow{h}_t &= \mathrm{LSTM}_{fwd}(f_t, \overrightarrow{h}_{t-1}), \\ \overleftarrow{h}_t &= \mathrm{LSTM}_{bwd}(f_t, \overleftarrow{h}_{t+1}), \\ a_t &= \mathrm{softmax}_t\left(w_a^\top [\overrightarrow{h}_t \oplus \overleftarrow{h}_t]\right), \\ y &= W_y \textstyle\sum_t a_t [\overrightarrow{h}_t \oplus \overleftarrow{h}_t] + b_y, \end{aligned} \tag{3}$$

where $\oplus$ is concatenation, a_t is a learned temporal attention weight that lets the network down-weight corrupted frames, and W_y, b_y are the parameters of the output layer.

V. Implementation

A. Phase 1: Data Ingestion

A preprocessing script indexes each database, performs the stratified split and caches the resized images as NumPy arrays. The full training set is held in memory and mini-batches are moved to the GPU, where a stochastic augmentation policy is applied per sample (horizontal/vertical flips, rotation $\pm 20°$, zoom $\pm 12\%$, shift $\pm 8\%$, brightness/contrast/saturation jitter $\pm 20\%$).

B. Phase 2: Training the Spatial Stages

The Stage-1 MobileNetV2 and the Stage-2 ResNet-50 are initialised from ImageNet and fine-tuned end-to-end with the Focal Loss

$$\mathrm{FL}(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t), \tag{4}$$

where p_t is the predicted probability of the ground-truth class, $\gamma = 2$ down-weights easy examples and α_t is a tempered inverse-frequency class weight ($\alpha_c \propto 1/\sqrt{n_c}$, normalised to mean 1). The loss is computed from log-softmax outputs for numerical stability. Optimisation uses AdamW [28] (weight decay 10^{-2}, decoupled from the gradient update), batch size 32 and a *differential learning rate*: 10^{-4} for the pretrained backbone and 10^{-3} for the new classifier head, to avoid catastrophic forgetting of ImageNet features. The learning rate follows a cosine annealing schedule [29] after a 10% linear warm-up:

$$\eta_t = \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min}) \left(1 + \cos\left(\frac{T_{cur}}{T_{max}}\pi\right)\right), \tag{5}$$

where T_{cur} is the current step and T_{max} the total number of steps. Because ImageNet-pretrained networks converge quickly on these data (Fig. 5), training runs for 5 epochs on Kvasir and 4 epochs on GastroVision; the checkpoint with the best validation macro-F1 is kept.

C. Phase 3: Training the Temporal Stage

With Stages 1–2 frozen, three synthetic clips (one on GastroVision, to save time) are generated for every training image (with augmentation, and each frame corrupted with probability 0.3), and their ResNet-50 features are cached. The Bi-LSTM is trained on these features for 40 epochs with focal loss, AdamW (lr 10^{-3}, cosine annealing) and random temporal reversal. Model selection uses an equal mix of clean and artifact-corrupted validation clips. Decoupling the expensive CNN from the recurrent model in this way reduces the temporal-training cost from hours to seconds.

Gate threshold. Missing a lesion is far more costly than running Stage 2 unnecessarily, so τ is chosen on the validation set as the *largest* threshold for which the gate still passes $\geq$99.5% of abnormal frames (Fig. 11). This gives $\tau = 0.27$ on Kvasir.

D. Phase 4: Evaluation and Explainability

Grad-CAM. The heat-map is computed from the gradients flowing into the last convolutional block of the Stage-2 ResNet-50 [19]. The importance weight of feature map A^k for class c is

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \tag{6}$$

and the weighted maps are combined and rectified so that only positively contributing features remain:

$$L^c_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right). \tag{7}$$

For display, L^c is up-sampled to the input resolution, normalised and scaled by the model's confidence for class c before being overlaid on the frame.

VI. Experimental Setup

A. Baselines

Nine baselines are compared with the proposed system: (i) a **classical** pipeline of dense SIFT [8] descriptors quantised into a 256-word bag of visual words [9], concatenated with an HSV colour histogram and a uniform LBP texture histogram [10], classified by an RBF-kernel SVM [11]; and (ii) eight ImageNet-pretrained deep networks fine-tuned with the *same* data, augmentation, optimiser, schedule and epoch budget as the proposed stages but with standard cross-entropy: AlexNet [12], VGG-16 (with batch normalisation) [13], GoogLeNet [14], ResNet-50 [16], DenseNet-121 [17], MobileNetV2 [23], EfficientNet-B0 [24] and ViT-B/16 [31].

B. Performance Evaluators

With TP, TN, FP, FN computed per class (one-vs-rest) and macro-averaged, we report: Accuracy; Precision $= TP/(TP+FP)$; Recall (sensitivity) $= TP/(TP+FN)$; F1 $= 2PR/(P+R)$; Specificity $= TN/(TN+FP)$; balanced accuracy (mean per-class recall); the area under the one-vs-rest ROC curve (AUC); Cohen's κ; and the Matthews Correlation Coefficient, which remains informative under class imbalance [34]:

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{D}}, \tag{8}$$

where $D = (TP+FP)(TP+FN)(TN+FP)(TN+FN)$; for $K > 2$ classes the multi-class generalisation computed from the full confusion matrix is used. Efficiency is measured by parameter count, GMACs per frame and batch-1 throughput (FPS) on the GPU after warm-up. Explainability is quantified against expert masks by the pointing-game hit rate [35], the fraction of CAM energy inside the mask and the IoU of the CAM thresholded at 0.5.

C. Test Protocols

Clean: the current frame is the unmodified test image (single-frame baselines see exactly the standard test set; the proposed system additionally sees four preceding synthetic frames). *Transient artifact*: only the current frame is corrupted with one of the four artifact types. *Artifact stream*: every frame of the clip is independently corrupted with probability 0.3. *Cross-database*: Kvasir-trained models are applied unchanged to Kvasir-SEG and CVC-ClinicDB.

VII. Results and Discussion

A. Baseline Comparison on Kvasir v2

Table III compares all methods on the Kvasir v2 test set with eight classification evaluators and three efficiency evaluators; Fig. 4 visualises accuracy, macro-F1 and MCC on both classification databases. The hand-crafted SIFT-BoVW+SVM pipeline reaches 77.25% accuracy, confirming that learned features are far better suited to the homogeneous, specular GI mucosa. Among the deep baselines, accuracy ranges from 87.58% (AlexNet) to 93.58% (ViT-B/16). The proposed cascade reaches 91.92% accuracy, 91.89% macro-F1, 90.79% MCC and 99.19% AUC. On clean, curated still images it is therefore 1.7 points below the best single-frame network and level with its own ResNet-50 backbone (91.92%); this is expected, since the temporal stage cannot add information when every frame is clean and the neighbouring frames are near-duplicates. The value of the design lies elsewhere: it is the only model that remains accurate when frames are degraded (Section VII-C), and a stronger backbone (e.g. ViT-B/16 or DenseNet-121) could be dropped into Stage 2 without changing the rest of the pipeline. The training curves in Fig. 5 show that every pretrained network converges within the 5-epoch budget, so the comparison is not biased by under-training.

Because the test sets contain about 1,200–1,600 images, accuracy differences of one or two points are within sampling noise. Table IV therefore reports bootstrap 95% confidence intervals and exact McNemar tests against the proposed system. On Kvasir the proposed cascade (91.92%, CI 90.2–93.4%) is significantly better than SIFT-BoVW+SVM, AlexNet, GoogLeNet, statistically indistinguishable from VGG-16, ResNet-50, DenseNet-121, MobileNetV2, EfficientNet-B0, and significantly behind only ViT-B/16.

The confusion matrices (Fig. 6) show that the dominant error on Kvasir is the confusion between *oesophagitis* and the *normal Z-line*, and between *dyed-lifted polyps* and *dyed resection margins*. Both pairs are visually near-identical and are known to be ambiguous even for annotators, which is also visible in the per-class F1 scores (Fig. 8). The ROC curves (Fig. 7) show micro-averaged AUCs above 0.99 for all deep models on Kvasir.

B. Generalisation to a Second Classification Database

Table V repeats the comparison on GastroVision, which covers more classes (including gastric findings) and is strongly imbalanced. Here plain accuracy is dominated by the large

TABLE III

BASELINE COMPARISON ON THE KVASIR V2 TEST SET (1,200 IMAGES, 8 CLASSES). PRECISION, RECALL, F1 AND SPECIFICITY ARE MACRO-AVERAGED; AUC IS MACRO ONE-VS-REST. BEST VALUE PER COLUMN IN BOLD. †EXPECTED PER-FRAME COST OF THE CASCADE GIVEN THE MEASURED GATE PASS RATE (GPU: APPLE M5, FP16; CPU: FP32).

	Classification (%)										FPS (batch 1)	
Model	Acc.	Prec.	Rec.	F1	Spec.	MCC	κ	AUC	Params (M)	GMACs	GPU	CPU
SIFT-BoVW+SVM	77.25	77.20	77.25	77.18	96.75	74.01	74.00	97.41	–	–	–	–
AlexNet	87.58	88.26	87.58	87.44	98.23	85.95	85.81	98.92	57.0	0.71	195	148.5
VGG-16	91.00	91.04	91.00	90.92	98.71	89.74	89.71	99.34	134.3	15.47	31	13.3
GoogLeNet	89.92	90.10	89.92	89.88	98.56	88.51	88.48	99.30	5.6	1.50	105	65.1
ResNet-50	91.92	92.08	91.92	91.89	98.85	90.79	90.76	99.57	23.5	4.09	54	37.0
DenseNet-121	92.75	92.87	92.75	92.73	98.96	91.74	91.71	99.55	7.0	2.83	40	27.2
MobileNetV2	90.58	90.76	90.58	90.56	98.65	89.27	89.24	99.30	2.2	0.30	127	19.8
EfficientNet-B0	91.58	91.61	91.58	91.57	98.80	90.39	90.38	99.38	4.0	0.38	113	3.2
ViT-B/16	**93.58**	**93.87**	**93.58**	**93.56**	**99.08**	**92.72**	**92.67**	**99.64**	85.8	11.27	51	12.5
Proposed cascade	91.92	92.04	91.92	91.89	98.85	90.79	90.76	99.19	28.4	3.35†	54†	13.9†

TABLE IV

STATISTICAL COMPARISON WITH THE PROPOSED CASCADE ON THE CLEAN TEST SETS: ACCURACY WITH A 2,000-SAMPLE BOOTSTRAP 95% CONFIDENCE INTERVAL AND THE EXACT MCNEMAR p-VALUE AGAINST THE PROPOSED SYSTEM ($^+/^-$: BASELINE SIGNIFICANTLY BETTER/WORSE AT $p < 0.05$).

	Kvasir v2		GastroVision	
Model	Acc. [95% CI]	p	Acc. [95% CI]	p
SIFT-BoVW+SVM	77.2 [74.8, 79.7]	$<0.001^-$	64.9 [62.3, 67.6]	$<0.001^-$
AlexNet	87.6 [85.6, 89.4]	$<0.001^-$	68.6 [65.9, 71.2]	$<0.001^-$
VGG-16	91.0 [89.3, 92.6]	0.278	74.9 [72.3, 77.3]	0.010^-
GoogLeNet	89.9 [88.2, 91.7]	0.017^-	76.3 [73.8, 78.6]	0.139
ResNet-50	91.9 [90.3, 93.5]	1.000	80.4 [78.2, 82.6]	0.035^+
DenseNet-121	92.8 [91.2, 94.2]	0.302	80.4 [78.0, 82.6]	0.058
MobileNetV2	90.6 [88.9, 92.2]	0.113	75.3 [72.8, 77.7]	0.021^-
EfficientNet-B0	91.6 [89.9, 93.2]	0.744	78.7 [76.3, 80.9]	0.718
ViT-B/16	93.6 [92.2, 94.9]	0.040^+	82.8 [80.6, 85.0]	$<0.001^+$
Proposed cascade	91.9 [90.2, 93.4]	–	78.2 [75.8, 80.5]	–

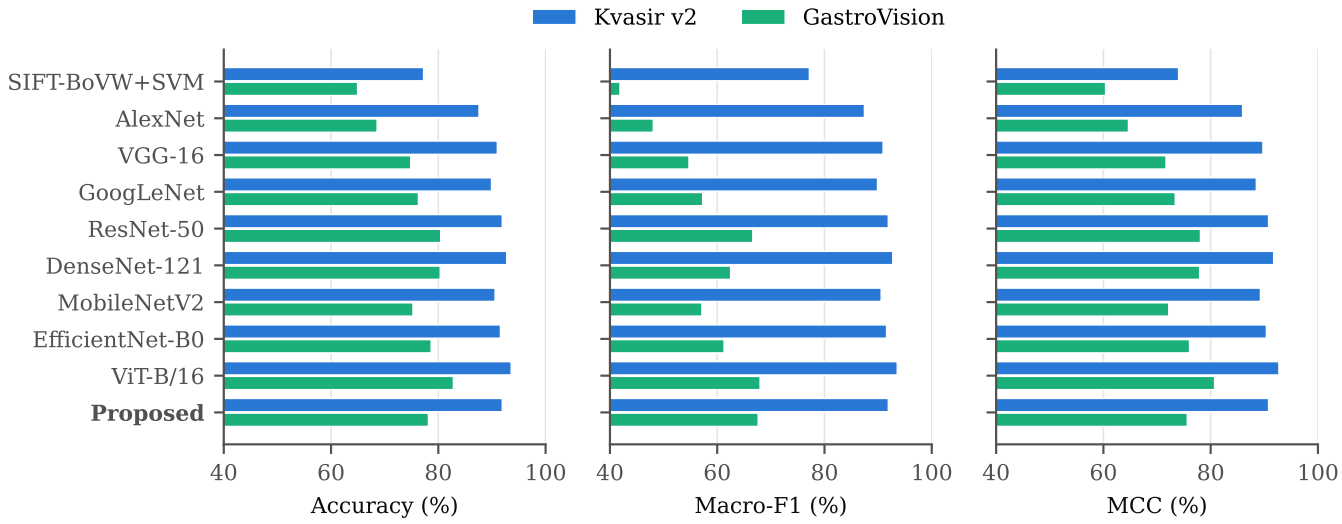


Fig. 4. Baseline comparison on the two classification databases (test sets). The proposed system is the bottom row of each panel.

classes, so balanced accuracy and MCC are the more informative evaluators. The proposed cascade obtains 78.15% accuracy and 67.55% balanced accuracy (best baseline: ViT-B/16, 82.81% / 66.29%). The classical baseline collapses on this database, especially in balanced accuracy, because its fixed descriptors cannot separate the many fine-grained classes. Although its plain accuracy is lower than that of the strongest single-frame networks, the proposed cascade has the highest

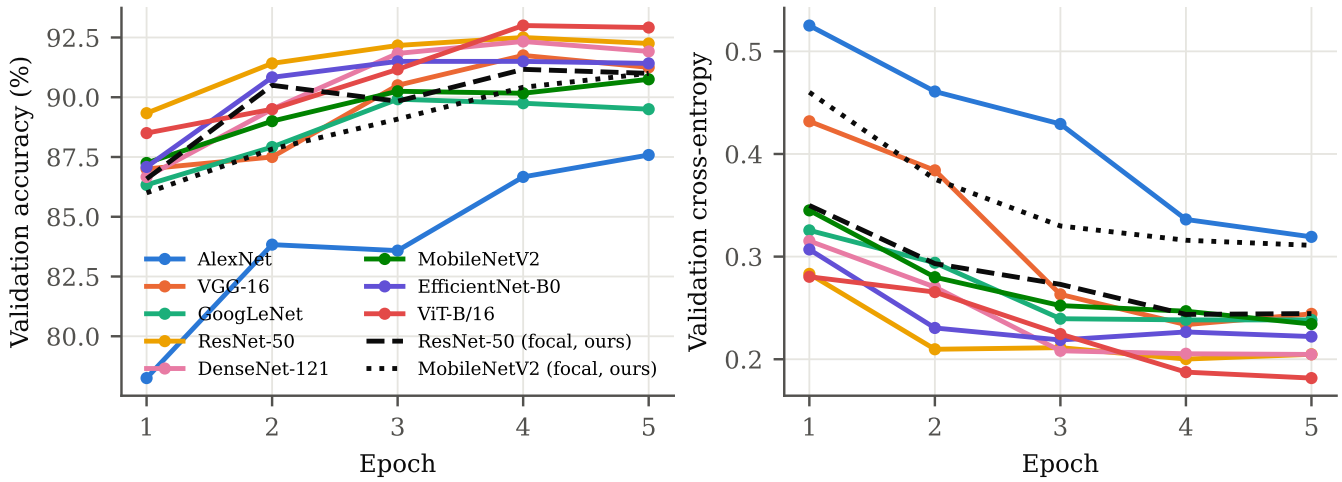


Fig. 5. Validation accuracy and cross-entropy per epoch on Kvasir v2 for all deep baselines and for the two focal-loss stages of the proposed system (black).

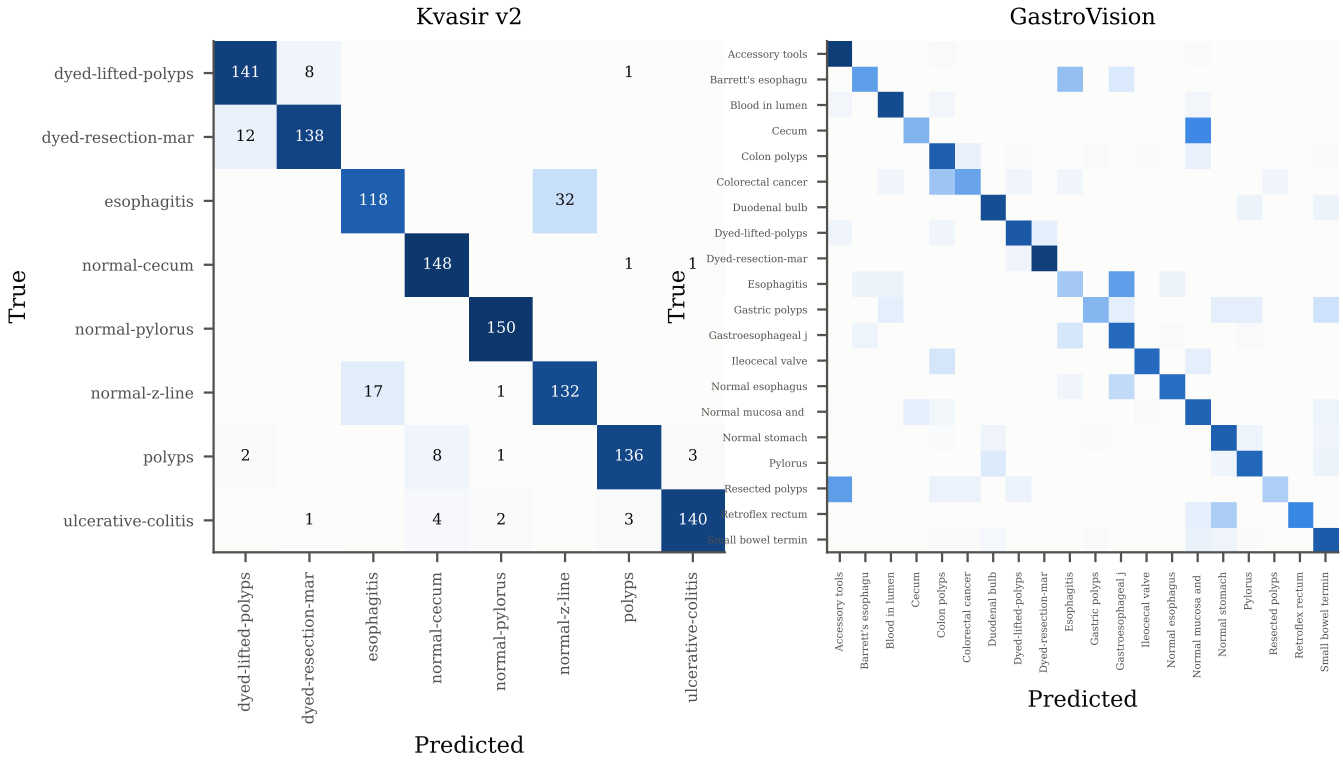


Fig. 6. Confusion matrices of the proposed cascade on the clean test sets (rows: true class).

balanced accuracy of all methods, i.e. the best recall on the rare classes, which is the intended effect of the focal loss and the abnormality-oriented gate.

C. Robustness to Endoscopic Artifacts

Table VI and Fig. 9 report Kvasir accuracy when the current frame is hit by a transient artifact. Every single-frame model degrades sharply, because it has nothing to classify except the corrupted frame: averaged over the four artifact types, the most robust baseline (ViT-B/16) keeps only 74.1%, and ResNet-50 keeps 54.2%. The proposed system keeps 91.8%, essentially its clean accuracy, and 91.2% on a random artifact stream where 30% of all frames are corrupted. This is the behaviour the temporal stage was designed for: a lesion that is visible in the preceding frames is not lost because of one blurred or flashed frame, and an artifact cannot create a lesion that the preceding frames do not show.

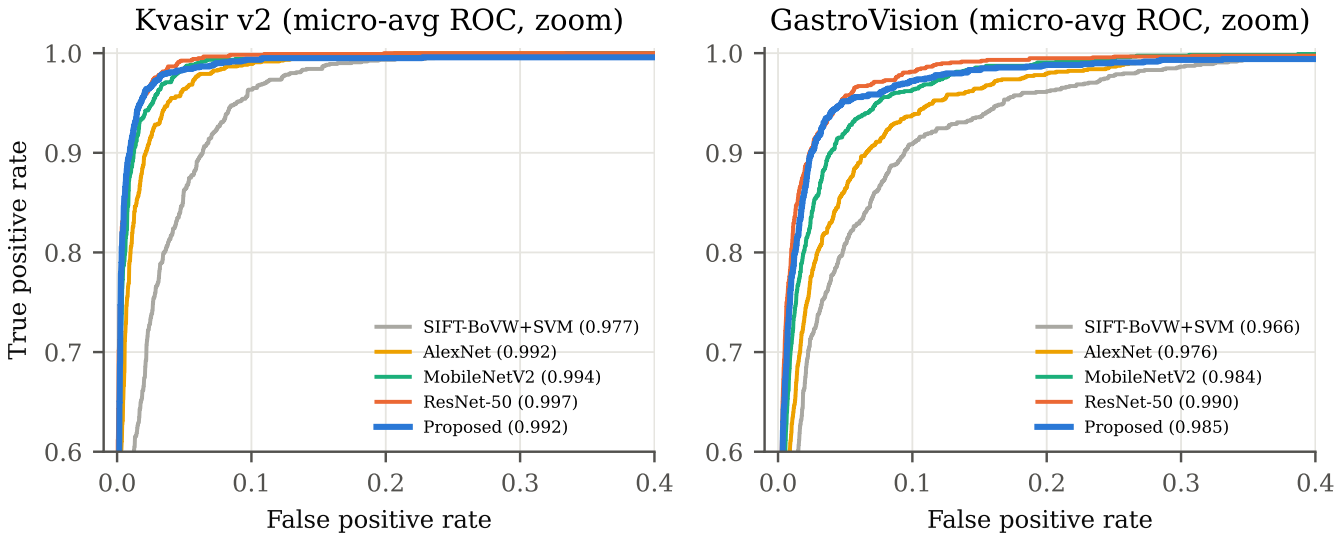


Fig. 7. Micro-averaged ROC curves (zoomed into the upper-left corner) with AUC in brackets.

TABLE V

GENERALISATION OF THE COMPARISON TO GASTROVISION (TEST SPLIT). B-ACC. = BALANCED ACCURACY (MEAN PER-CLASS RECALL), WHICH EXPOSES FAILURES ON RARE CLASSES THAT PLAIN ACCURACY HIDES.

	GastroVision (20 classes)			
Model	Acc.	B-Acc.	F1	MCC
SIFT-BoVW+SVM	64.94	42.19	41.86	60.40
AlexNet	68.59	48.79	48.08	64.67
VGG-16	74.85	53.50	54.73	71.67
GoogLeNet	76.29	56.35	57.28	73.39
ResNet-50	80.44	64.99	66.64	78.09
DenseNet-121	80.36	60.69	62.50	77.94
MobileNetV2	75.28	56.32	57.14	72.18
EfficientNet-B0	78.66	61.01	61.27	76.05
ViT-B/16	**82.81**	66.29	**67.99**	**80.73**
Proposed cascade	78.15	**67.55**	67.63	75.63

TABLE VI

KVASIR TEST ACCURACY (%) WHEN THE *CURRENT* FRAME IS HIT BY A TRANSIENT ARTIFACT (COLUMNS 2–5), AND ON A RANDOM ARTIFACT STREAM ($p = 0.3$ PER FRAME). SINGLE-FRAME MODELS SEE ONLY THE CURRENT FRAME; THE PROPOSED SYSTEM ADDITIONALLY SEES THE FOUR PRECEDING FRAMES.

Model	Clean	Motion blur	Defocus	Specular flash	Noise/occlusion	Stream	Mean art.
AlexNet	87.6	75.9	64.2	39.2	42.3	79.2	55.4
VGG-16	91.0	67.3	65.9	43.0	29.6	79.8	51.5
GoogLeNet	89.9	52.6	63.3	46.8	40.8	80.1	50.9
ResNet-50	91.9	69.7	55.2	47.2	44.5	83.2	54.2
DenseNet-121	92.8	64.4	50.7	47.5	47.8	82.4	52.6
MobileNetV2	90.6	67.8	46.4	43.2	22.9	78.0	45.1
EfficientNet-B0	91.6	72.9	67.2	48.5	11.1	81.2	49.9
ViT-B/16	93.6	84.1	74.5	72.4	65.2	88.2	74.1
Proposed	**91.9**	**91.3**	**92.1**	**91.2**	**92.3**	**91.2**	**91.8**

Fig. 10 shows why: the learned temporal attention a_t assigns corrupted frames an average weight an order of magnitude lower than clean frames at every position of the window, i.e. the Bi-LSTM has learned *which* frames to trust without any frame-quality supervision.

D. Ablation Study

Table VII isolates each component. (i) Replacing cross-entropy by focal loss changes clean Kvasir accuracy only marginally on this balanced database. (ii) Any form of temporal aggregation removes most of the artifact sensitivity of the single-frame model. Simple mean pooling of the per-

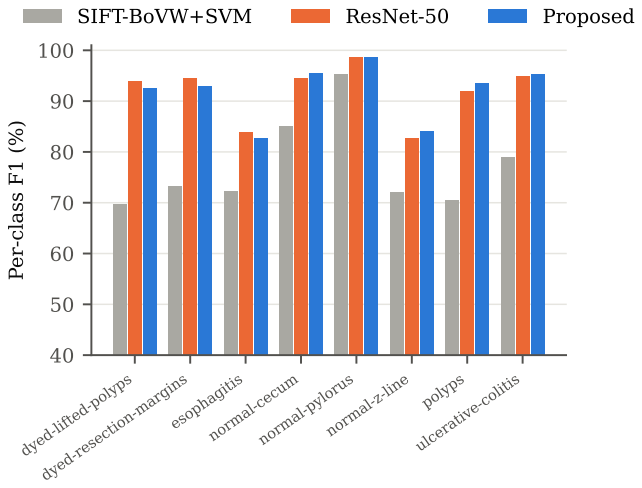


Fig. 8. Per-class F1 on Kvasir v2.

TABLE VII
ABLATION OF THE PROPOSED COMPONENTS. "ARTIFACT STREAM": EVERY FRAME OF THE 5-FRAME WINDOW IS INDEPENDENTLY CORRUPTED WITH PROBABILITY 0.3.

Configuration	Clean Acc.	Clean F1	Clean MCC	Artifact stream Acc.	Artifact stream MCC
Kvasir v2					
ResNet-50 (CE), single frame	91.92	91.89	90.79	83.25	80.97
ResNet-50 (focal), single frame	91.42	91.35	90.27	81.83	79.34
+ temporal mean pooling	92.33	92.31	91.25	91.83	90.69
+ Bi-LSTM (no gate)	91.92	91.89	90.79	91.17	89.93
+ gate on current frame only	91.83	91.81	90.69	90.42	89.07
Proposed cascade	91.92	91.89	90.79	91.17	89.93
GastroVision					
ResNet-50 (CE), single frame	80.44	66.64	78.09	71.89	68.36
ResNet-50 (focal), single frame	77.82	66.95	75.40	69.01	65.38
+ temporal mean pooling	80.02	69.99	77.79	81.03	78.81
+ Bi-LSTM (no gate)	78.32	67.46	75.79	79.76	77.35
+ gate on current frame only	77.39	66.95	74.76	78.66	76.12
Proposed cascade	78.15	67.63	75.63	79.59	77.15

frame probabilities is a strong baseline on our synthetic clips, because neighbouring frames are near-duplicates; the Bi-LSTM performs comparably while additionally providing interpretable attention weights and the capacity to model genuine appearance changes in real video. (iii) Routing on the *current* frame only makes the gate itself the weak point: a flash or a dark occlusion can push a lesion frame below τ and cause a bypass. Averaging the gate probabilities over the five-frame buffer (gate scores are computed once per frame anyway) removes this failure mode at no extra cost, and the full cascade matches the un-gated temporal model while skipping Stages 2–3 for part of the stream.

E. Computational Efficiency and the Stage-1 Gate

The gate threshold is selected on validation data as the largest τ that keeps the abnormal-frame recall at $\geq$99.5% (Fig. 11); this gives $\tau = 0.27$ on Kvasir, a test-set gate recall of 99.3% for abnormal frames and a bypass rate of 25.3% of all

TABLE VIII
STAGE-1 GATE ON THE CLEAN TEST SETS: NORMAL/ABNORMAL CLASS SPLIT, SELECTED THRESHOLD τ, RECALL OF ABNORMAL FRAMES, SPECIFICITY ON NORMAL FRAMES, FRACTION OF FRAMES THAT BYPASS STAGES 2–3, AND EXPECTED CASCADE GPU FPS (BATCH 1).

Database	Norm./Abn.	τ	Recall	Spec.	Bypass	FPS
Kvasir v2	3/5	0.27	99.3	66.4	25.3	54
GastroVision	10/10	0.06	99.4	29.7	18.1	51

frames (Table VIII). Measured on the idle M5 GPU at batch size 1, the gate costs 7.9 ms and Stages 2–3 cost 14.4 ms per frame, so the expected cost of the cascade at Kvasir's class mix is 18.6 ms (54 FPS) and 3.35 GMACs, against 18.6 ms and 4.09 GMACs for ResNet-50 on every frame: at this prevalence the cascade roughly breaks even, while adding the temporal verification for free. On the CPU, PyTorch's depthwise-convolution kernels for this ARM processor are inefficient (MobileNetV2 is slower than ResNet-50 despite needing 13$\times$ fewer MACs), so the gate brings no CPU wall-clock gain without an optimised mobile runtime; CPU figures are nevertheless reported in Table III.

The benefit of the cascade grows with the fraction of normal frames in the stream (Fig. 12). Kvasir is a curated, class-balanced benchmark in which only 3 of 8 classes are normal, whereas most frames of a full screening examination show normal mucosa. With 80% normal frames the expected cost falls to 2.21 GMACs (69 FPS on the GPU), and with 90% normal frames to 1.94 GMACs (73 FPS), compared with 4.09 GMACs / 54 FPS for ResNet-50 and 11.27 GMACs / 51 FPS for ViT-B/16 applied to every frame. Fig. 13 places all models in the accuracy–throughput plane.

F. Cross-Database Generalisation

Table IX applies the Kvasir-trained models, without any retraining, to the two pixel-annotated polyp databases. Kvasir-SEG images are drawn from the same source as Kvasir, so its numbers mainly measure in-domain consistency; 991 of 1000 images remain after removing near-duplicates of training images. CVC-ClinicDB comes from a different hospital and endoscope and is therefore a true domain-shift test. On CVC-ClinicDB the proposed cascade flags 93.5% of the polyp frames as abnormal (80.1% as *polyp*). The gap between the polyp-class and abnormal-class sensitivities shows that many shifted-domain polyps are still routed to the lesion branch even when the fine-grained label is wrong, which is the clinically relevant behaviour for a detection aid. ViT-B/16 and EfficientNet-B0 transfer best to the new domain, which suggests that the Stage-2 backbone, rather than the temporal design, limits cross-domain robustness; on the in-domain Kvasir-SEG images the proposed cascade has the highest polyp sensitivity (98.5%).

G. Explainability

Grad-CAM maps of the Stage-2 network (Fig. 14) are compared quantitatively with the expert polyp masks in Table X.

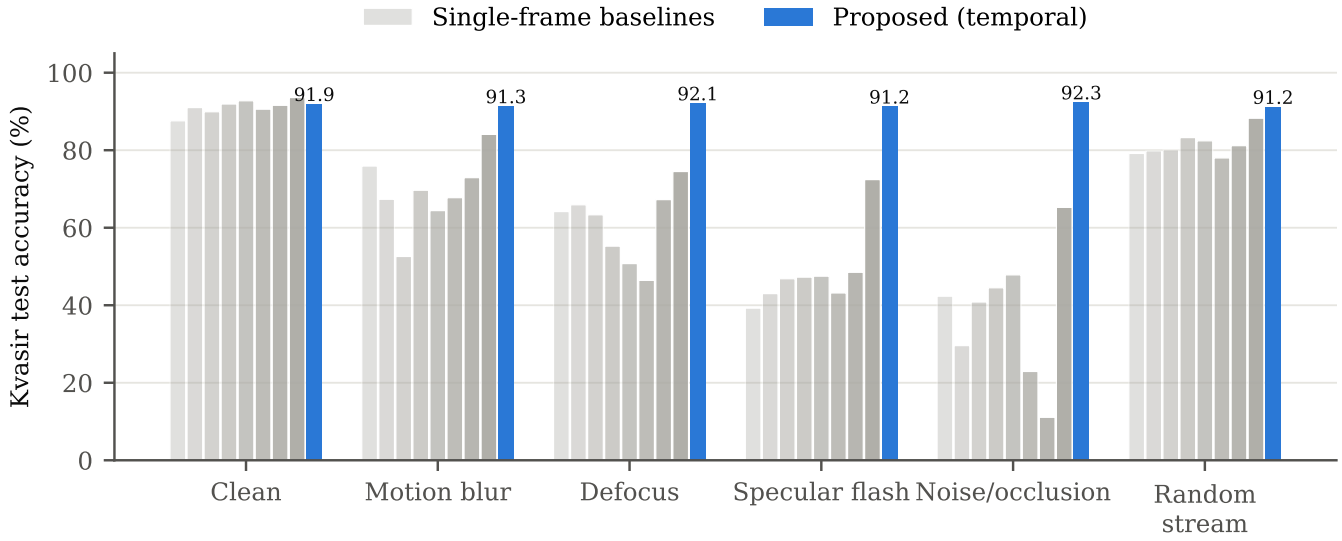


Fig. 9. Kvasir test accuracy under transient artifacts on the current frame and on a random artifact stream. Grey bars: the eight single-frame deep baselines (left to right as in Table III); blue: proposed.

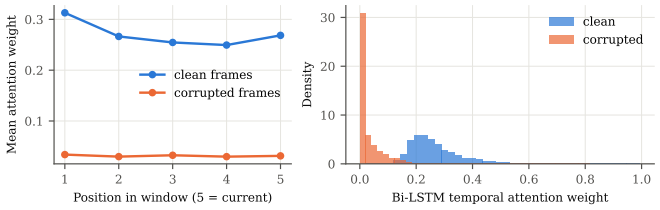


Fig. 10. Bi-LSTM temporal attention on the artifact stream: mean weight per window position (left) and the distribution of weights (right) for clean versus corrupted frames.

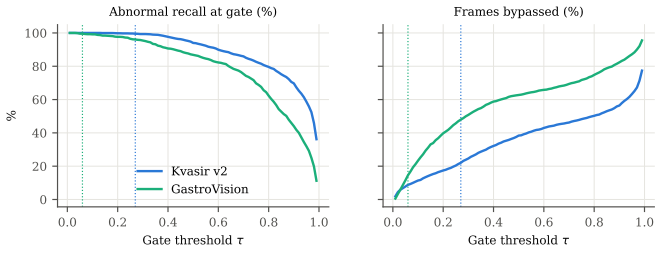


Fig. 11. Gate threshold sweep on the validation sets: recall of abnormal frames and fraction of bypassed frames. Dotted lines mark the selected τ.

TABLE IX

CROSS-DATABASE GENERALISATION OF KVASIR-TRAINED MODELS ON UNSEEN POLYP DATABASES: SENSITIVITY (%) FOR THE *POLYP* CLASS AND FOR *ANY ABNORMAL* CLASS. KVASIR-SEG IMAGES DUPLICATED IN THE KVASIR TRAINING SPLIT WERE REMOVED.

Model	Kvasir-SEG (n=991) Polyp	Kvasir-SEG (n=991) Abnormal	CVC-ClinicDB (n=612) Polyp	CVC-ClinicDB (n=612) Abnormal
AlexNet	88.2	95.3	69.8	92.0
VGG-16	96.4	98.7	85.6	97.4
GoogLeNet	96.3	98.6	77.1	91.0
ResNet-50	97.4	99.2	78.8	92.8
DenseNet-121	95.2	98.2	74.7	91.7
MobileNetV2	96.6	98.7	74.3	92.6
EfficientNet-B0	97.4	98.7	84.0	97.7
ViT-B/16	96.6	99.7	91.3	96.6
MobileNetV2 (focal)	96.3	99.0	75.2	95.6
ResNet-50 (focal)	97.0	99.1	79.4	94.9
Proposed cascade	98.5	99.4	80.1	93.5

TABLE X

QUANTITATIVE GRAD-CAM LOCALISATION AGAINST EXPERT POLYP MASKS (400 IMAGES PER DATABASE): POINTING-GAME HIT RATE (PG), FRACTION OF CAM ENERGY INSIDE THE MASK (EIM) AND IOU OF THE CAM THRESHOLDED AT 0.5.

Model	Kvasir-SEG PG	Kvasir-SEG EiM	Kvasir-SEG IoU	CVC-ClinicDB PG	CVC-ClinicDB EiM	CVC-ClinicDB IoU
ResNet-50 (CE)	86.8	45.8	42.7	67.8	36.8	33.6
ResNet-50 (focal, proposed Stage 2)	81.2	42.2	38.1	65.0	34.1	31.4
DenseNet-121 (CE)	81.8	31.7	41.0	58.0	24.4	31.1
MobileNetV2 (focal, Stage 1)	70.0	38.9	40.7	53.5	26.6	29.8

The pointing-game hit rate of the proposed Stage-2 network is 81.2% on Kvasir-SEG and 65.0% on CVC-ClinicDB, against a random-pointing baseline equal to the mean mask area (14.5% and 8.5%). This indicates that the network attends to the lesion itself rather than to dataset artifacts such as the green position-indicator box or text overlays. Interestingly, the cross-entropy ResNet-50 localises slightly better than the focal-loss version on both databases: focal loss down-weights easy, confidently classified polyps, which yields lower-confidence and somewhat more diffuse maps; this trade-off between rare-class sensitivity and map sharpness is worth further study. Finally, the t-SNE embedding [36] of the Stage-2 features (Fig. 15) shows compact class clusters, with the expected overlap between oesophagitis and Z-line and between the two dyed classes.

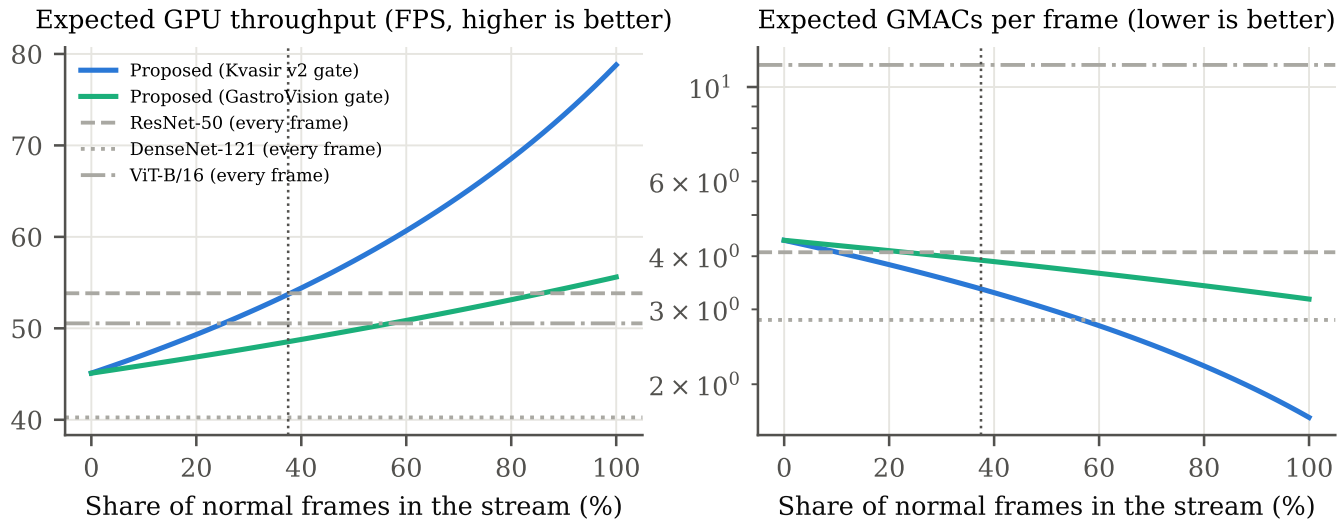


Fig. 12. Expected cascade cost as a function of the share of normal frames, computed from the measured component latencies and the gate's measured recall/specificity: GPU throughput (left) and GMACs per frame (right, log scale). Grey lines are single-frame baselines run on every frame; the dotted line marks Kvasir's test prevalence.

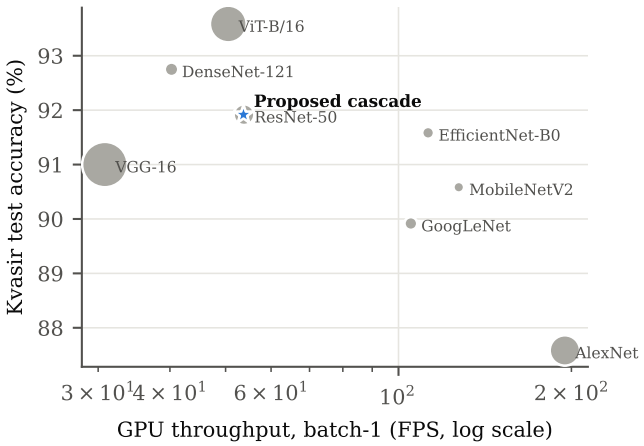


Fig. 13. Accuracy versus batch-1 GPU throughput on Kvasir v2 (marker area $\propto$ parameters).

H. Limitations

The temporal clips are synthesised from still images, so the study measures robustness to *simulated* artifacts; evaluating on real annotated video (e.g. the HyperKvasir video collection [2]) is the most important next step. All networks were fine-tuned for a short, fixed budget on a single train/validation/test split; repeated runs with different seeds would provide confidence intervals. Rare classes with fewer than 30 images, including gastric ulcers in GastroVision, had to be excluded, which also limits the strength of the claims for rare gastric pathologies.

VIII. Conclusion

This paper presented a cascaded spatio-temporal framework for GI abnormality detection in endoscopy that combines a MobileNetV2 gate, a ResNet-50 feature extractor and a Bi-LSTM with temporal attention, trained with focal loss and explained with Grad-CAM. It was benchmarked against a classical SIFT-BoVW+SVM pipeline and eight deep baselines on four public databases with classification, efficiency and explainability evaluators. On clean still images the proposed system comes within 1.7 points of the best single-frame network (91.92% accuracy and 90.79% MCC on Kvasir v2, versus 93.58% for ViT-B/16). Its advantage appears when frames are degraded: under transient motion blur, defocus, specular flashes or occlusion it keeps 91.8% accuracy, whereas single-frame baselines fall to 74.1% or less. A temporally smoothed gate retains $\geq$99% of abnormal frames while skipping the expensive stages for normal ones, which cuts the average compute per frame and pays off increasingly as the share of normal frames grows, as in real examinations. Future work will validate the system on real endoscopic video, add repeated-seed confidence intervals, and extend the label set to rare gastric pathologies such as early gastric cancer and ulcers.

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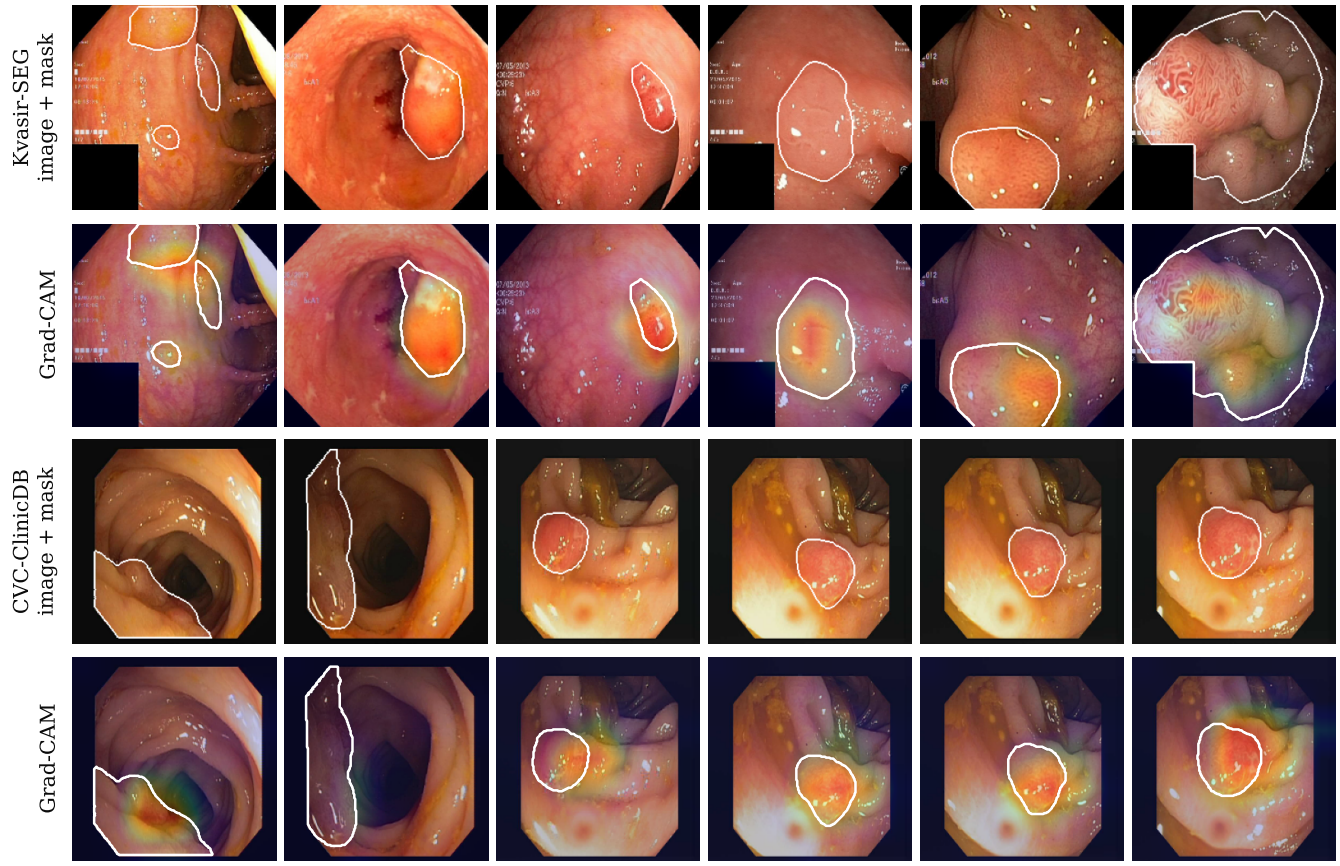


Fig. 14. Grad-CAM of the proposed Stage-2 network for the *polyp* class on unseen Kvasir-SEG and CVC-ClinicDB frames; the expert polyp mask is outlined in white. The map is scaled by the model's confidence.

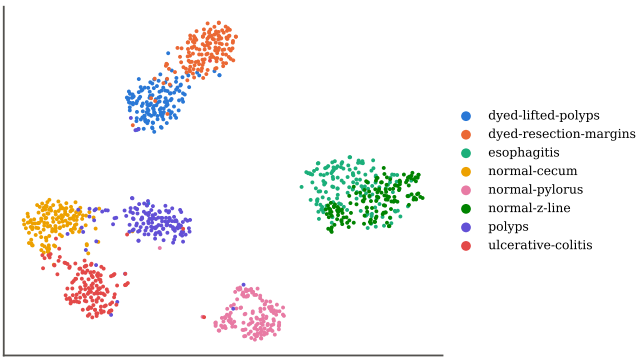


Fig. 15. t-SNE of the 2048-d Stage-2 features of the Kvasir test images.

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